

WHAT CAN THE LOCAL GPS NETWORK TELL US ABOUT THE GLACIAL ISOSTATIC ADJUSTMENT OF LONG ISLAND, NY?

Moore, M.¹, Brown, R.¹, Thomas, M.¹, Santella, K.¹, Ballato, B.¹, Marsellos, A.E.¹, Tsakiri, K.G.²

¹Dept. Geology, Environment and Sustainability, Hofstra University, NY, USA,

²Department of Information Systems, Analytics, and Supply Chain Management, Rider University, NJ, USA



Introduction

Objective: observe if last glacial melting causes isostatic rebound of Long Island by analyzing continuous GNSS (GPS) observations using R programming language and RStudio software

Comparing other works:

- There are multiple scientific papers on the isostatic rebound and glacial isostatic adjustment of other areas (Bogusz et al., 2019), (George et al., 2011), and
- Glacial Isostatic Adjustment (GIA) models were made for Northern America (e.g. Sella et al., 2007),
- but none specifically focused on local regions like Long Island, NY

Performing spectral analysis and identifying trends:

- Spectral analysis: Allows through a periodogram to identify the appropriate window size for a moving average application
- Moving Average: It removes noise and it increases the signal-to-noise ratio
- A spectral analysis prior to the moving average application provides best results

Methodology: The GPS Data

- Data: Nevada Geodetic Laboratory's Global GPS Network: Sites NYCI, NYQN, NYRH, and ZNY1
- Lollipop: Using ggplot2 → site selection (time frame of data, missing data, etc)

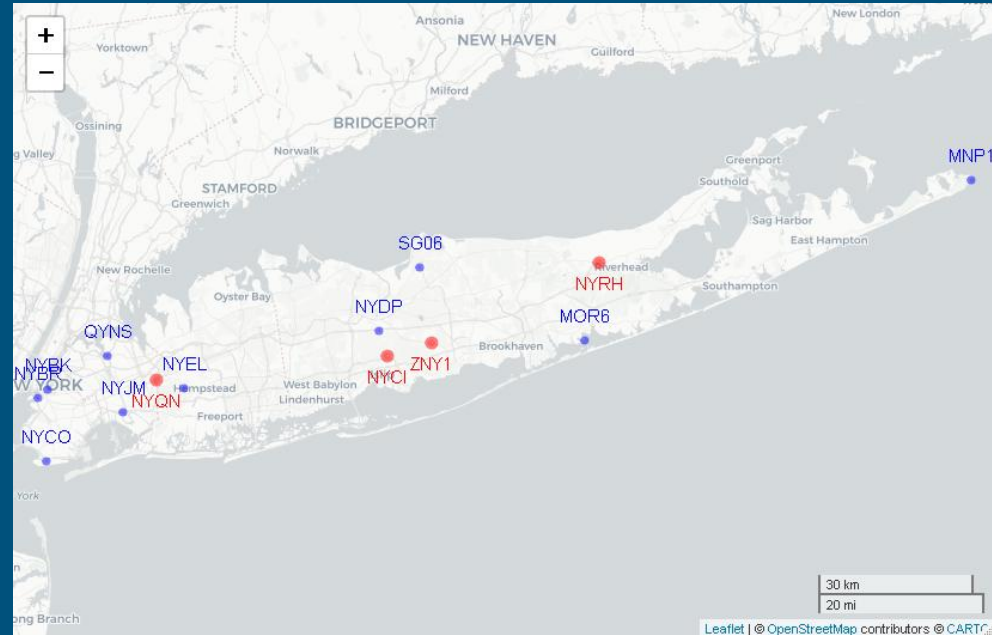
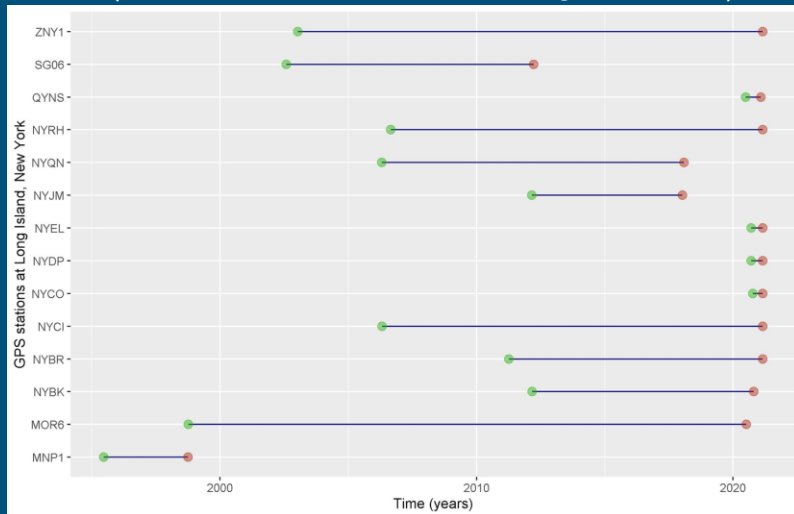


Figure 1 (Above): Map of GPS sites on Long Island, NY (the red represents the GPS sites that were used in our research while the blue are the sites that had a shorter span of data)

Figure 2 (Left): Span of all GPS sites' data in Long Island, NY

Methodology cont.

- All figures and data sets as well as the equations were created through R programming language (Rstudio)
- Created data set of the northward, eastward, and upward movement of all four GPS sites
 - Outliers removed with z-score (-3,3) function in R

$$Z = \frac{x - \mu}{\sigma}$$

Expression 1: Formula to find the z-score

```
data2_No_Outliers <-  
  data2 %>%  
  group_by(site) %>%  
  mutate(zscore = (up - mean(up))/sd(up)) %>%  
  filter(!(zscore < -3)) %>%  
  filter(!(zscore > 3))
```

Figure 3: Code that removed outliers in R using the z-score function

Methodology cont.

- Spectral analysis (SP) applied on raw data using periodogram function
 - Identify a suitable window size for moving average (MA)
- Moving average application was performed by `kz` function in R
 - Removed high frequencies based on results of SP

Methodology cont.

- Correlation matrix was created to observe possible connections between northward, eastward, and upward movement

```
matrix <- data2 %>%  
  select(-2) %>%  
  group_by(site) %>%  
  do(data.frame(Cor=t(cor(.[,2:4], .[,2:4]))))  
  
matrix$ENU <- c("east", "north", "up", "east", "north", "up",  
               "east", "north", "up", "east", "north", "up")  
matrix <- matrix %>% select(1,5,2,3,4)  
matrix
```

Figure 4: Code that created a correlation matrix out of the data from all four GPS sites

Results

Spectral analysis *(using TSA: periodogram function in R)*

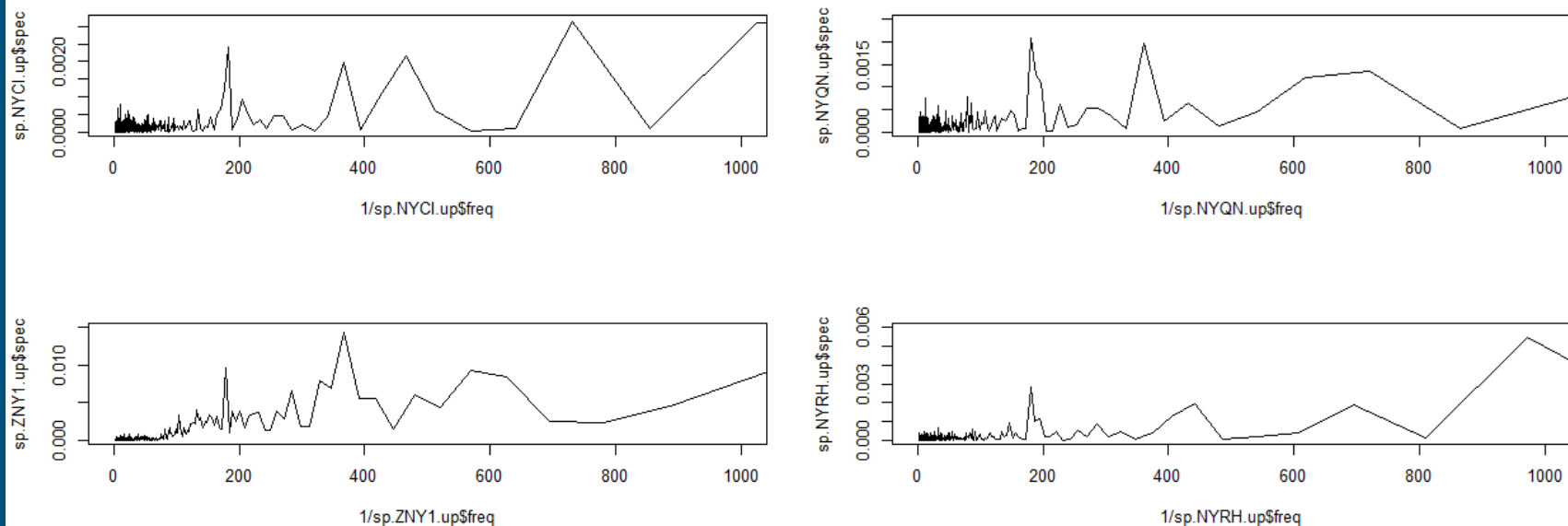


Figure 3: Spectral analysis of the selected GPS sites (horizontal axis is the period in days)

Results

Spectral analysis *(using TSA: periodogram function in R)*

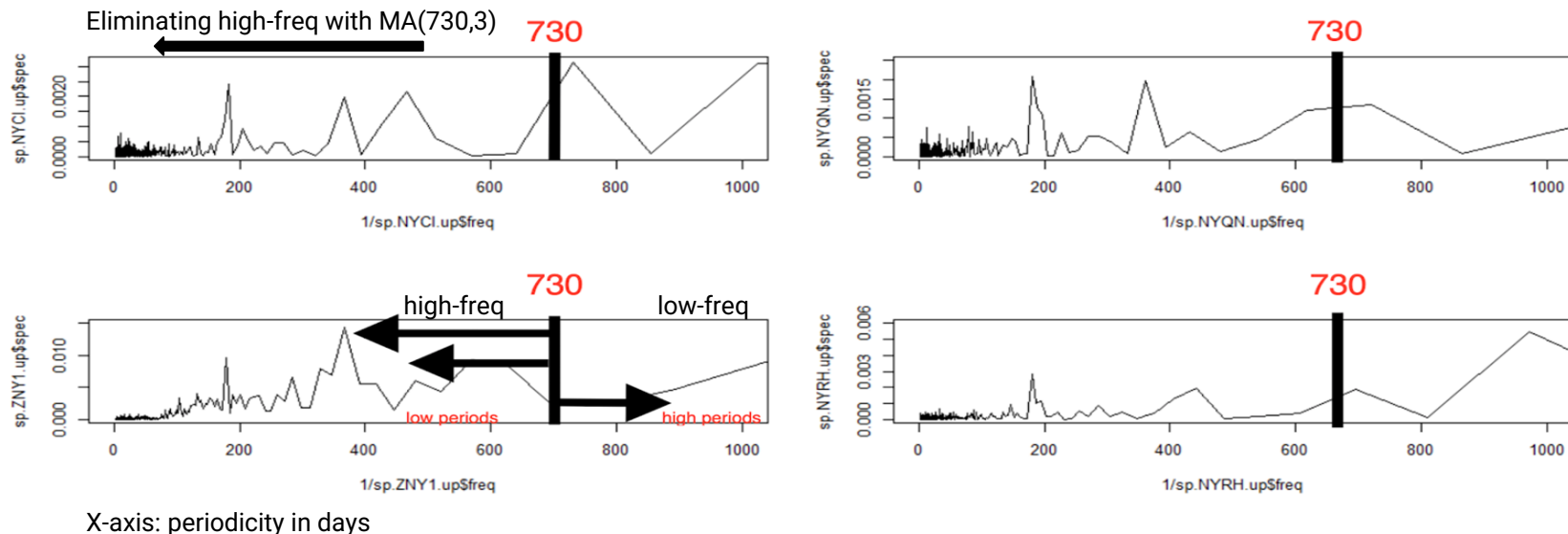


Figure 3: Spectral analysis of the selected GPS sites (horizontal axis is the period in days)

Results cont.

- Uniform behavior of a continuous subsidence of Long Island, NY from the vertical components of the GPS sites.
 - 0.01 mm downtrend per decade.
- No presence of uplift related to glacial isostatic rebound.

Figure 3: Raw data of displacements (in mm) from the selected GPS sites that include outliers

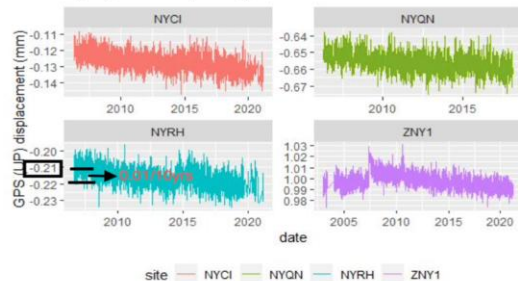


Figure 4: Raw data of displacements (in mm) from the selected GPS sites in which outliers have been excluded using a z-score (-3,3).

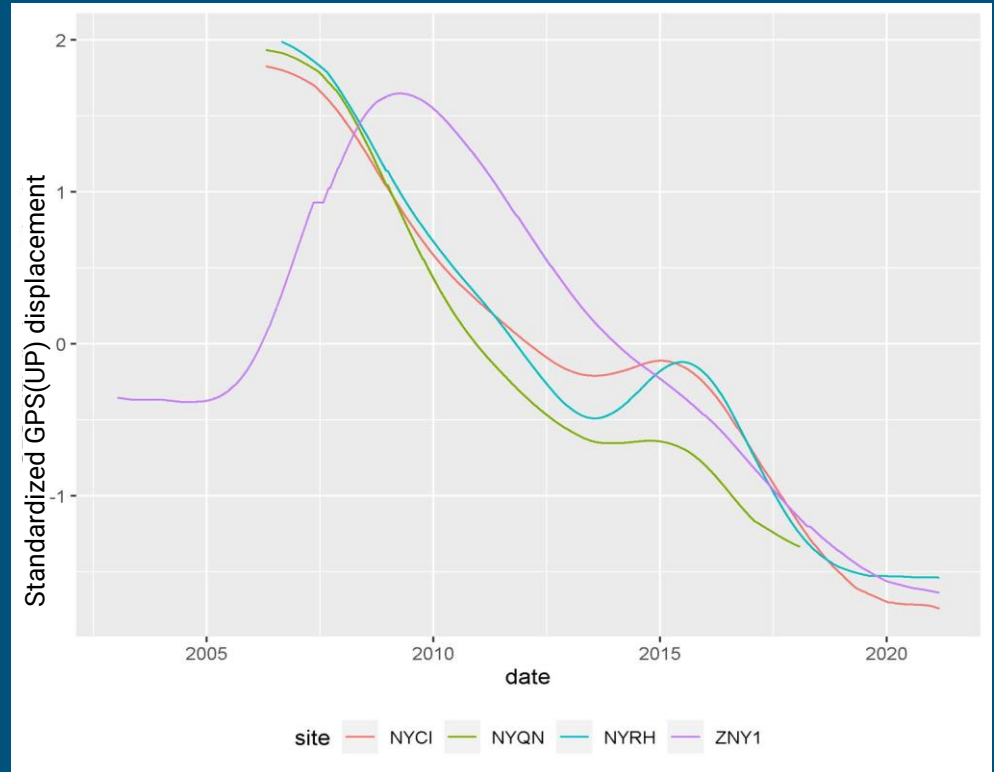


Figure 4: Displacements (up) from the GPS selected sites NYCI, NYQN, NYRH, and ZNY1 after being averaged by the $kz(730,3)$ function moving average. Data were standardized to fit at the same scale

Discussion

Why monitor the glacial isostatic adjustment?

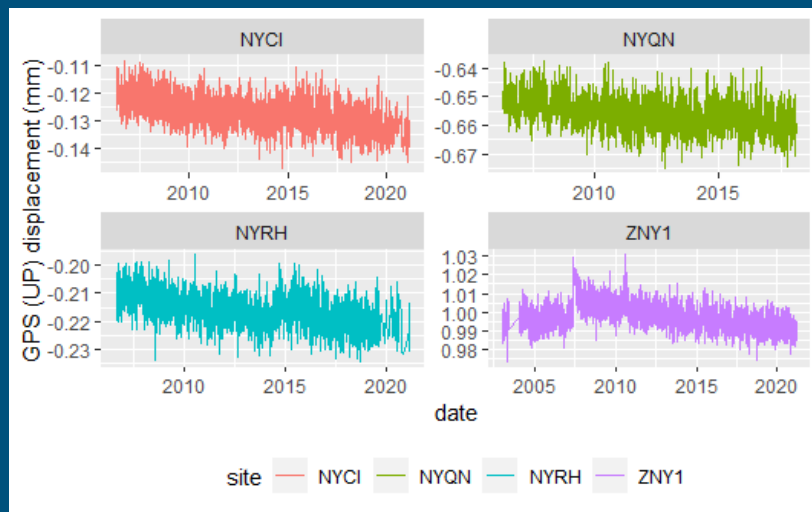
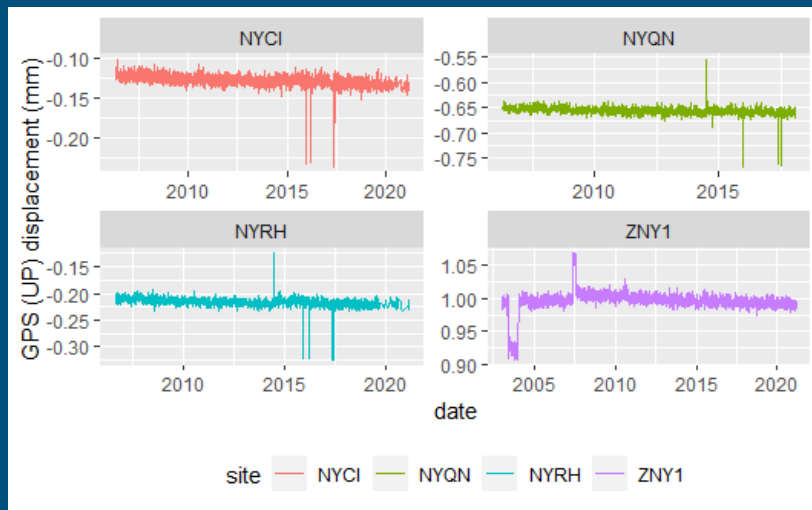
- Long term trends provide us a glimpse into the future. Isostatic adjustment can cause changes in Earth's gravitational field, rotation rate, and may cause earthquakes.
- Utilizing GPS coordinates allows us to notice any significant patterns. However, we have not proceeded to further decomposition of higher frequencies and related short term patterns.
- One important observation is the weak association of the up component with both east and north GPS components. Potential causes of this association is due to atmospheric or other geologic phenomena. E.G (fault reactivation, local ground deformation)

Discussion

With Outliers-->

What is the overall impact of removing outliers?

- Outliers can dramatically affect the statistical results of a study. The mean of a data set can be strongly influenced by an outlier.
- Any change in the mean could lead to an inaccurate interpretation.
- Unknown phenomenon shown in figure three could be caused by earthquakes or GPS modifications. E.G. (antenna replacement)



Without Outliers-->

Discussion

- Our research has primarily focused on long-term trends, resulting in a moving average of a large window which is more practical.
- The implications of the study indicates that Long Island's glacial isostatic adjustment has not applied during the GPS logged interval.

Conclusion

- GPS (GNSS) continuous measurements show evidence of a consistent downtrend connected to the subsidence of Long Island with no relation to glacial isostatic adjustment.
 - Substantial evidence showing a period of uplift during a 3 year interval of 2013-2015.
- In the long term it does not appear that Long Island will be affected by significant subsidence or uplift. Instead, Long Island is affected by sea level change.

References

- Bogusz, J, Klos, A, Pokonieczny, K, 2019. Optimal Strategy of a GPS Position Time Series Analysis for Post-Glacial Rebound Investigation in Europe, *Remote Sensing* 11, no. 10: 1209.
- Close, B.; Zurbenko, I. Kolmogorov-Zurbenko Adaptive Filters. Available online: <http://cran.r-project.org/web/packages/kza/index.html>. March 16, 2021.
- George, N.V., Tiampo, K.F., Sahu, S.S., Mazzotti, S., Mansinha, L., Panda, G., 2012. Identification of Glacial Isostatic Adjustment in Eastern Canada Using S Transform Filtering of GPS Observations, *Pure and Applied Geophysics*, 169 (2012), pp. 1507-1517.
- Marsellos, A.E., Hantzis, K.D., Tsakiri, K.G., 2020. How to remove the seasonal atmospheric effect: An application on a surface volcanic temperature at Nisyros active volcano, Hellenic Volcanic Arc, *Journal of Volcanology and Geothermal Research*, Volume 407.
- Sella, G., Stein, S., Wdowinski, S., Dixon, T., Dokka, R., 2004, GPS observation of glacial isostatic adjustment in North America, *Cah. Cent. Eur. Geod. Seismol.*, 23, 105– 110.
- Sella, G., Stein, S., Dixon, T., Craymer, M., James, T., Mazzotti, S., Dokka, R., 2007. Observation of glacial isostatic adjustment in “stable” North America with GPS, *Geophysical Research Letters*, Volume 34, Issue 2.
- Tsakiri, K.; Zurbenko, I.G., 2013. Explanation of fluctuations in water use time series. *Environ. Ecol Stat.* 20, 399–412.
- Tsakiri, K.G.; Zurbenko, I.G., 2011. Prediction of ozone concentrations using atmospheric variables. *Air Qual. Atmos. Health*, 4, 111–120.
- Wikipedia. Kolmogorov-Zurbenko Filters. Available online: http://en.wikipedia.org/wiki/Kolmogorov%E2%80%93Zurbenko_filter. March 16, 2021.
- Yang, W., Zurbenko, I., kzf: Kolmogorov–Zurbenko Fourier transform and application. R package, 2006.
- Zurbenko, I.G., M. Sowizral, 1999. Resolution of the destructive effect of noise on linear regression of two time series, *Far East J. Theor. Stat.*, 3, pp. 139-157.
- Zurbenko, I.G. *The Spectral Analysis of Time Series*; Amsterdam North Holland Series in Statistics and Probability; Elsevier North-Holland: New York, NY, USA, 1986.

Credit authorship contribution statement

Moore, M.: R coding, conceptualization, writing- Methodology plus part of References and Results, editing, data curation and analysis; Brown, R.: Abstract writing, results writing, conceptualization; Thomas, M.: R coding, editing, data analysis, writing- Discussion, part of References; Santella, K.: R coding, data curation and analysis, conceptualization, writing - Introduction, part of Abstract, part of Results, part of References, editing; Ballato, B.: conceptualization, writing, editing, part of abstract, part of introduction, part of conclusion, R coding; Marsellos, A.: Supervision, Conceptualization, R coding, editing, writing; Tsakiri, K.G.: R coding, writing - Methodology